Building an Enterprise Identity Infrastructure with Active Directory Domain Services (AD DS)

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Introduction

Modern organizations rely on Active Directory Domain Services (AD DS) to centrally manage users, computers, authentication, and network resources. Instead of configuring every computer individually, organizations use a Domain Controller to provide centralized identity management and security.

Objective

The objective of this task is to install and configure Active Directory Domain Services, promote the Windows Server to a Domain Controller, create a new Active Directory domain, join a Windows 10 client to the domain, and perform basic Active Directory administration.

Theory Topics

Active Directory

It is a database and set of services that connect users with the network resources they need to get their work done.

Active Directory Domain Services (AD DS)

Stores information about users, computers, and other devices on the network, helps administrators securely manage this information and facilitates resource sharing and collaboration between users.

Benefits of Active Directory

- Simplified Resource Sharing

- Streamlined User Management: providing a centralized platform to create, modify or delete users across the entire network.
- Enhanced Network Security: Group policies and access controls enforce strict password requirements and limit users' access to specific files or applications based on their specific roles within the company.
- Scalability: Active Directory can accommodate large networks with thousands of users, computers, and other objects.
- Fault Tolerance: Active Directory supports features like replication and failover, which ensure fault tolerance of directory services.

Components & Architecture of Active Directory

- Domain: the core organizational units in Active Directory that define administrative boundaries within a network, typically associated with an individual, company, or organization.
- Domain controller: servers that manage security authentication requests, enforce security policies, and replicate directory information across the network.
- OU: containers for managing and organizing entities like computers, groups, and users. They assist in assigning administrative duties and enforcing group rules.
- Objects: are entities that represent resources that are present in the AD network.
- Groups: a way to collect user accounts, computer accounts, and other groups into manageable units.
- LDAP (Lightweight Directory Access Protocol) : protocol used by Active Directory to access and manage directory information. It provides a standardized way for services to interact with the directory service.

- Kerberos Authentication: Kerberos is an authentication protocol that is used to verify the identity of a user or host.

  "A Kerberos is a system or router that provides a gateway between users and the internet. Therefore, it helps prevent cyber attackers from entering a private network. It is a server, referred to as an "intermediary" because it goes between end-users and the web pages they visit online".

- SYSVOL Folder: The system volume (SYSVOL) is a special directory on each DC. It is made up of several folders with one being shared and referred to as the SYSVOL share.
- NTDS Database: it is the database for (AD DS) which stores directory data and makes that data available to network users and administrators, it also includes password hashes and user details for all computer and user objects in the domain.
- Global Catalog: a distributed data repository that contains partial copies of all objects in the Active Directory forest. It enables searches for objects across domains.
- Flexible Single Master Operations (FSMO) roles: five special Active Directory tasks assigned to specific domain controllers to prevent data conflicts.

  1. Schema Master

  2. Domain Naming Master

  3. Relative ID (RID) Master

  4. Primary Domain Controller (PDC) Emulator

  5. Infrastructure Master (domain level)

Centralized Identity Management

It is the operating model in which identity records, roles, policy, and access decisions are governed through a single authoritative layer rather than being replicated independently across each application

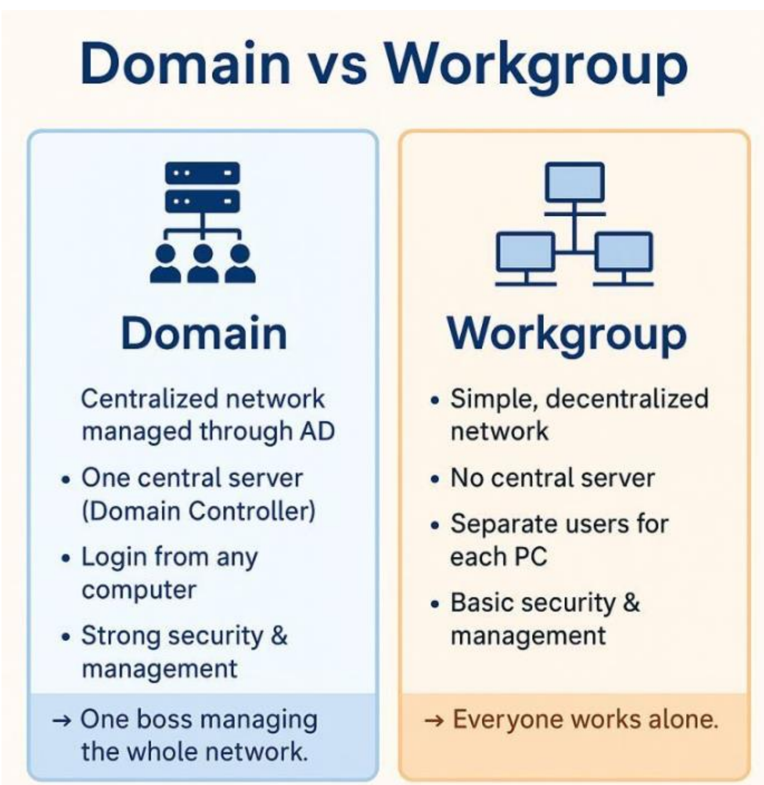
In this model IAM all happens in one environment. In a workplace setting, this looks like the user signing into a single workspace to access all the applications and tools they need.

Authentication vs Authorization

Authentication (identity verification): Determines whether users are who they claim to be.

Authorization (access control determination): Determines what users can and cannot access.

Workgroup vs Domain



Domain Functional Level

It is an advanced features that determines the minimum Windows Server version allowed for domain controllers.

Why DNS is Required for AD DS

To make it possible for clients to locate domain controllers and for the domain controllers that host the directory service to communicate with each other.

Software Requirements

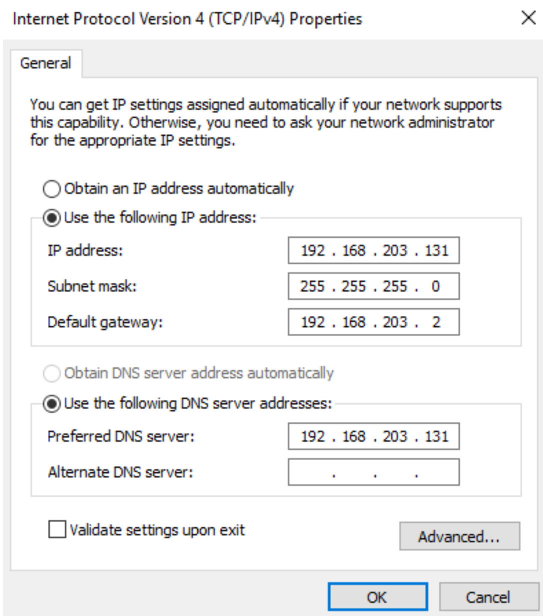
Software: VM workstation
Environment1: Windows Server 2019
Environment2: Windows 10 Client VMs

Prerequisites: DNS & DHCP configured
Access Level: Administrator Account

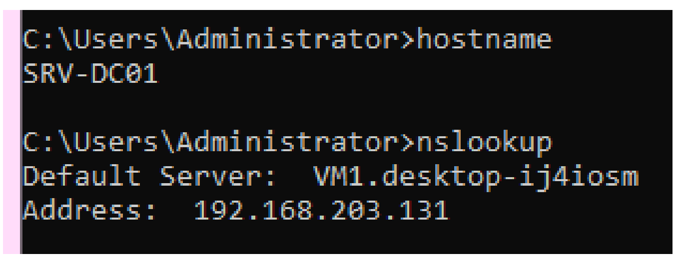
Implementation

1. Verifying that the Windows Server is ready for Active Directory deployment.

Static IP Address



DNS & hostname

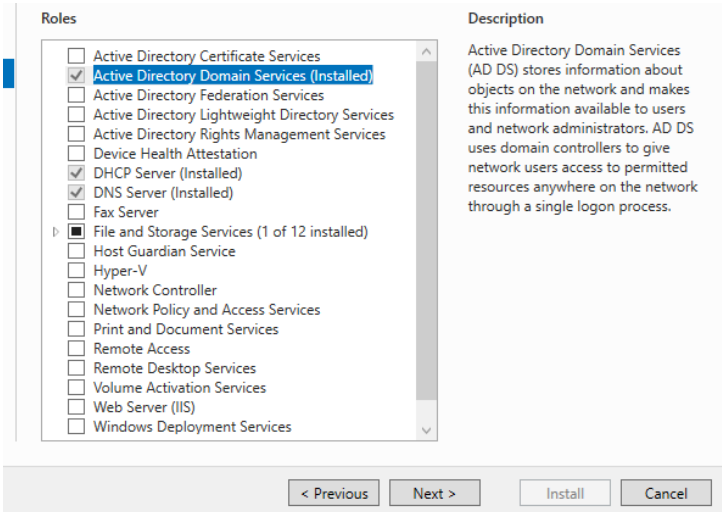


```
C:\Users\Administrator>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : SRV-DC01
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 00-0C-29-10-4D-32
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::204c:8130:88bf:b759%10(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.203.131(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.203.2
   DHCPv6 IAID . . . . . . . . . . . : 83889193
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-E8-49-04-00-0C-29-10-4D-32
   DNS Servers . . . . . . . . . . . : 192.168.203.131
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

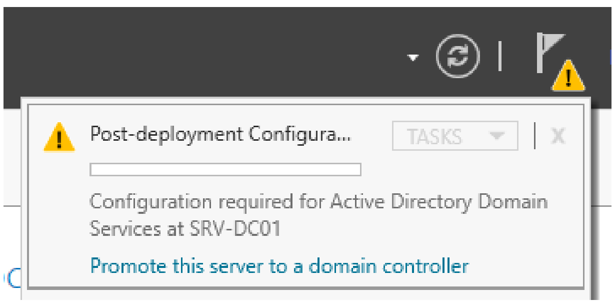
connectivity

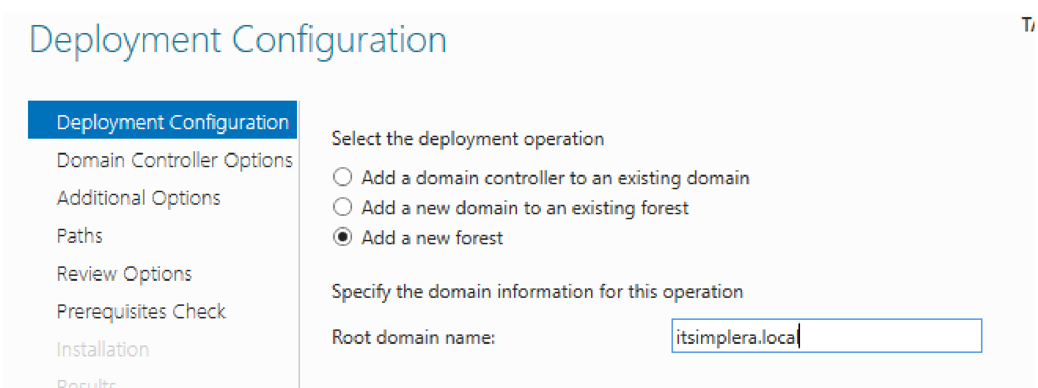
```
C:\Users\Administrator>ping 192.168.203.131

Pinging 192.168.203.131 with 32 bytes of data:
Reply from 192.168.203.131: bytes=32 time<1ms TTL=128
Reply from 192.168.203.131: bytes=32 time<1ms TTL=128
Reply from 192.168.203.131: bytes=32 time<1ms TTL=128
Reply from 192.168.203.131: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.203.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

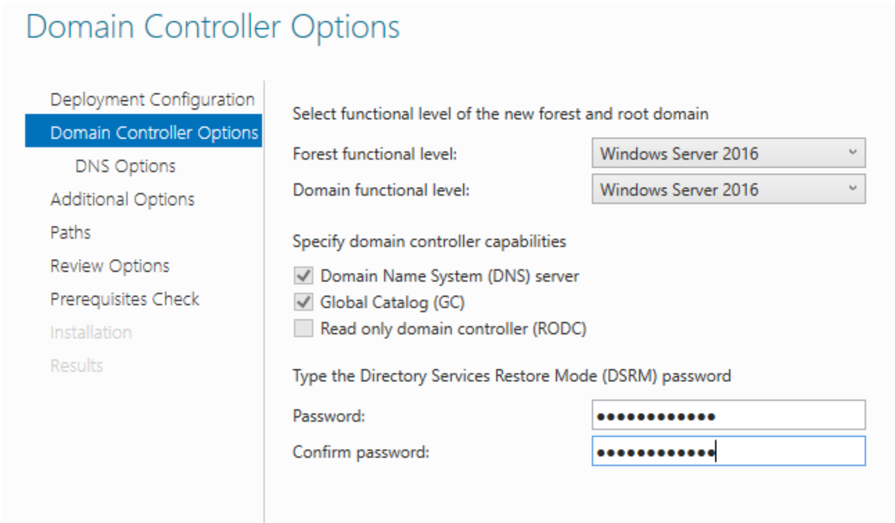
2. Installing the Active Directory Domain Services role.



3. Promoting the Server to a Domain Controller



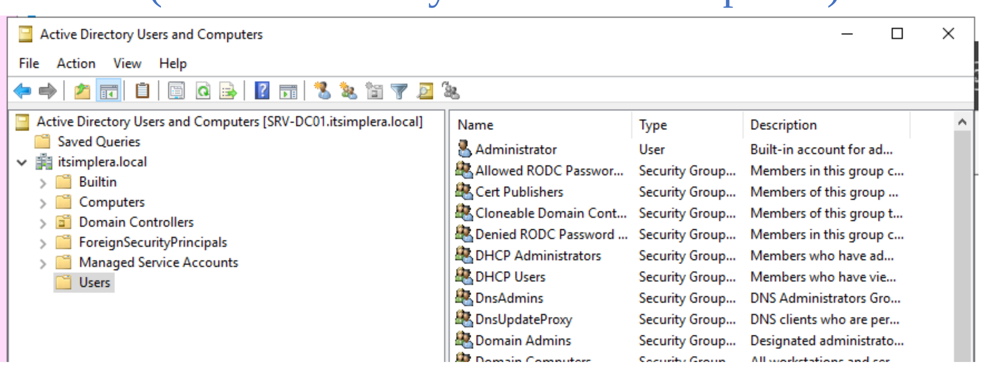




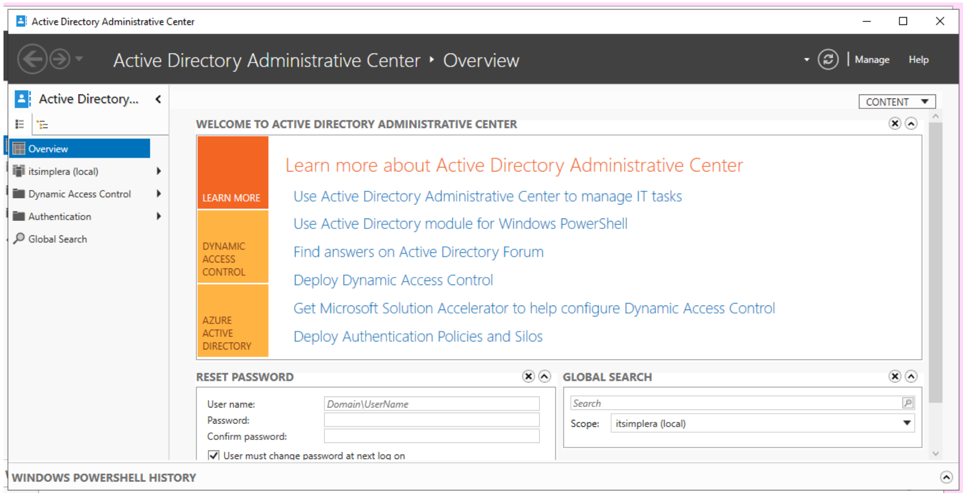
Legacy Compatibility Name: NetBIOS is an older networking protocol. Windows uses this short name (typically the first part of your DNS domain name, such as `ITSIMPLERA` for `itsimplera.local`) to support legacy applications, older operating systems, and user login formats like `ITSIMPLERA\username`.

4. Active Directory Management Tools

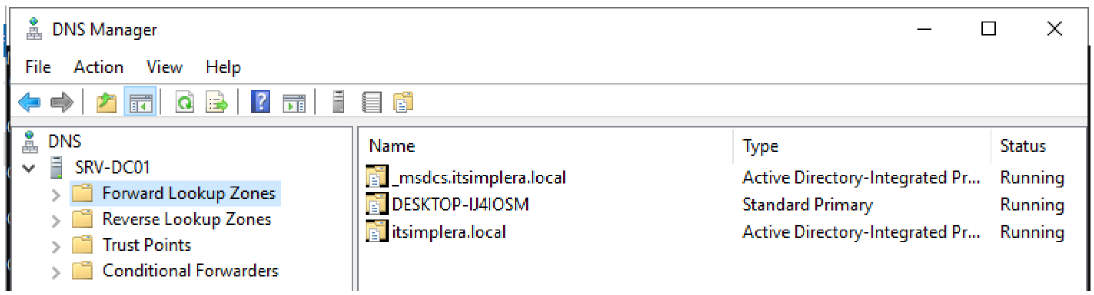
dsa.msc (Active Directory Users and Computers)



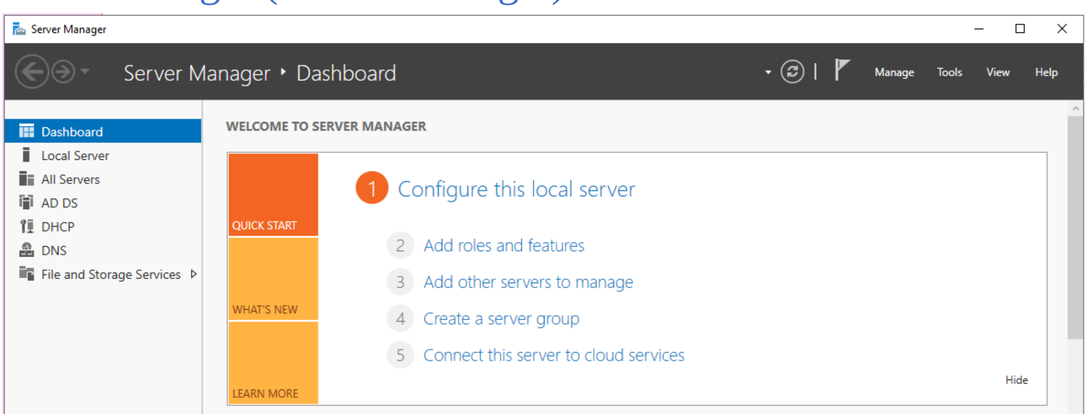
dsac.exe (Active Directory Administrative Center)



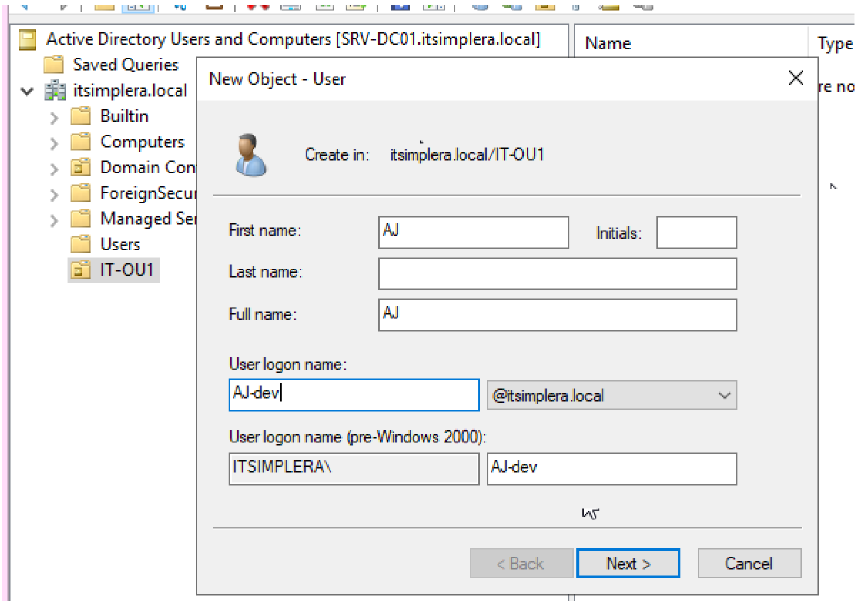
dnsmgmt.msc (DNS Manager)

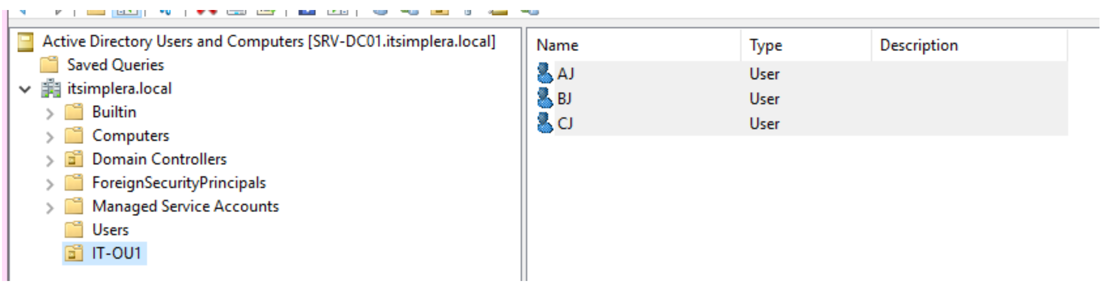


servermanager (Server Manager)

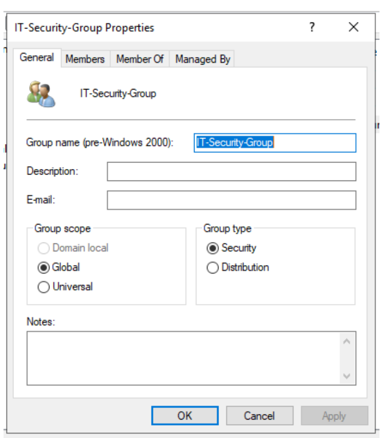


5. Creating Active Directory Objects
   * One Organizational Unit (OU) -> active directory users & computers -> right-click on domain -> new -> OU
   * Domain User Accounts -> right-click on OU -> new -> user

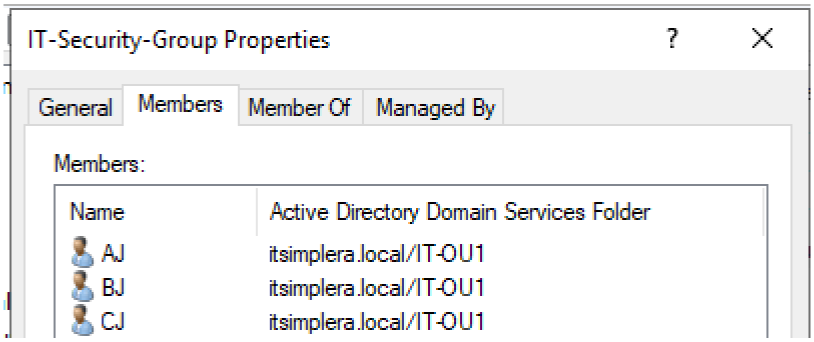




* Security Group -> on OU empty space -> right-click -> new -> group



* Adding users to the Security Group -> double click on security group -> properties -> members -> add



* Resetting a user's password -> right-click on user -> reset password

* Disabling a user's account -> right-click on user -> disable account

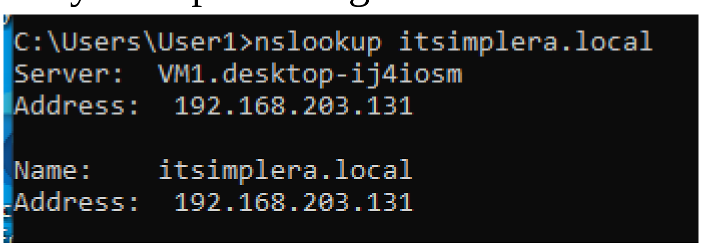
* Enabling a user’s account -> right-click on user -> enable account

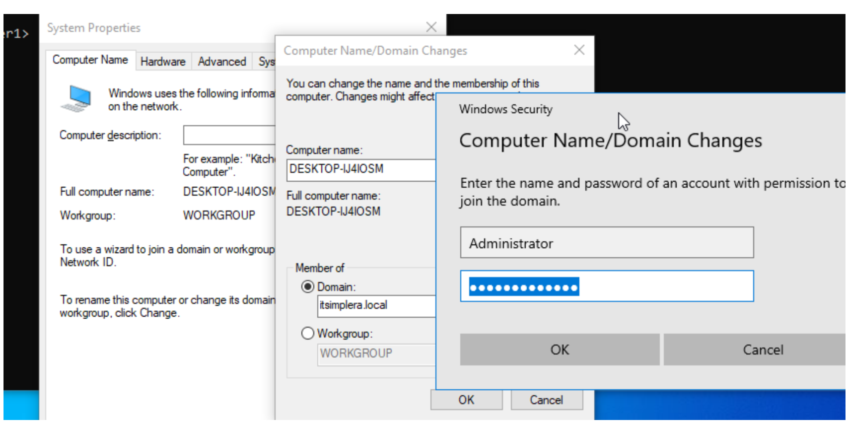
* Renaming a user’s account -> right-click on user -> rename

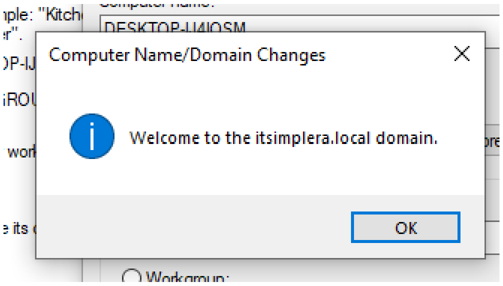
6. Joining a Windows 10 Client to the Domain

   -> Configuring the Preferred DNS Server to the Domain Controller

   -> sysdm.cpl – change – select domain – administrator credentials







7. Verifying Active Directory Functionality

Domain Controller is operational

```
C:\Users\Administrator>echo %logonserver%
\\SRV-DC01
```

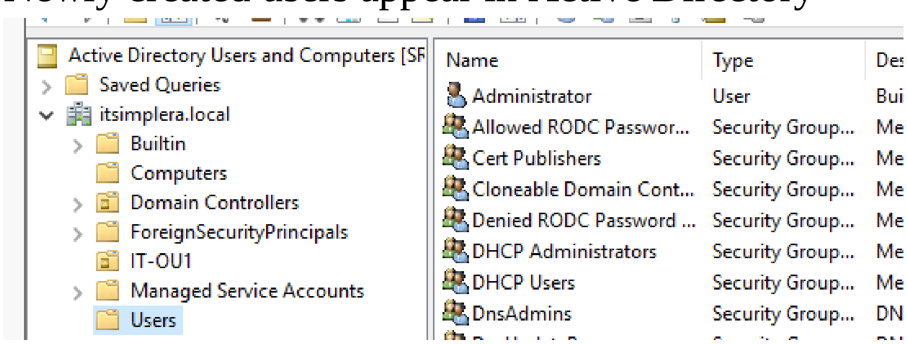
DNS Service is functioning

```
C:\Users\Administrator>nslookup
Default Server:  VM1.desktop-ij4iosm
Address:  192.168.203.131
```

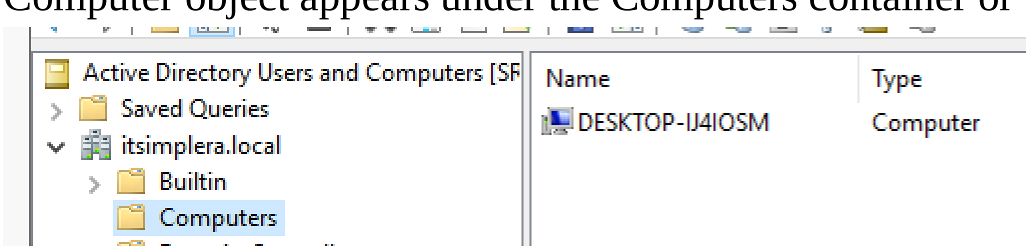
Client successfully authenticates to the domain

```
C:\Users\Administrator>whoami
itsimplera\administrator
```

Newly created users appear in Active Directory



Computer object appears under the Computers container or OU



Challenges faced

1. Problem: A client cannot join the domain

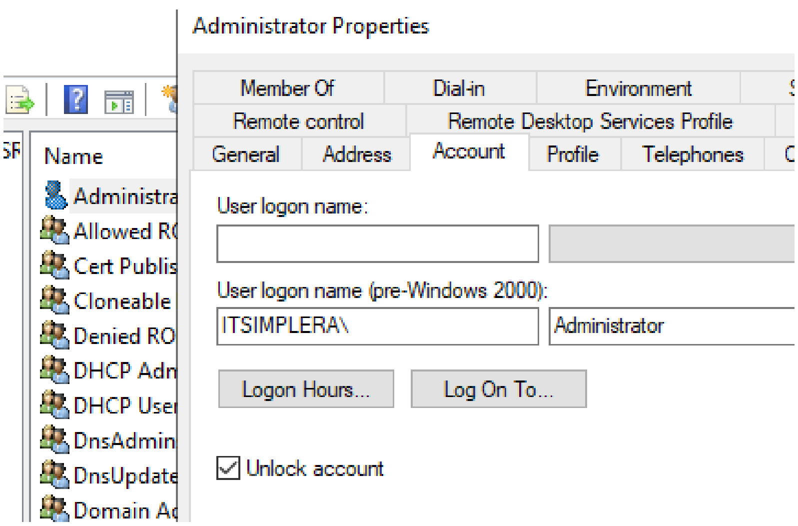
   Cause: Preferred DNS Server was set to the wrong IP address

   Solution: Set correct Preferred DNS to Server IP address

2. A user cannot log in using domain credentials

   Cause: The user account is either locked out due to multiple incorrect password attempts, or the password has expired/is flagged to be changed at next logon

   Solution: Unlock the account in ADUC



Learning outcomes

How to:

- Manage users, computers, Organizational Units (OUs), and security groups within Active Directory Domain Services (AD DS).
- Promote a Windows Server to a Domain Controller and create a new Active Directory forest and domain.
- Join a Windows 10 client to an Active Directory domain and verify successful domain authentication.
- Perform common administrative tasks, including creating user accounts, managing group memberships, resetting passwords, and enabling or disabling accounts.
- Understand the role of DNS in Active Directory and verify proper DNS integration.
- Troubleshoot basic Active Directory issues, including domain join failures, authentication problems, and DNS resolution errors.

Conclusion

Understanding Active Directory Domain Services (AD DS) is essential for managing enterprise networks through centralized authentication, user management, and resource access. The concepts and practical activities completed in this task provided valuable experience in deploying and administering a domain environment while reinforcing the importance of DNS integration, domain management, and basic Active Directory troubleshooting.

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